Example 1: Training Program

A fitness program records resting heart rate before and

after for 12 participants. Differences (before − after) have
mean $\bar{d}$ = 6.1 bpm and s_d = 4.0 bpm. Test $\alpha = 0.05$ (right-

tailed) to see if rates decreased.

- $t = 6.1 / (4.0/\sqrt{12}) \approx 5.28$, df = 11.
- Critical $t = 1.796 \Rightarrow$ reject H_0.
- 95% CI: $6.1 \pm 2.201 \cdot 4.0/\sqrt{12} \Rightarrow (3.6, 8.6)$ bpm.
- $d = 6.1 / 4.0 = 1.53$ (large effect).
- Conclusion: Program significantly reduces resting heart

rate.

Example 2: Productivity Tool (Two-Tailed)

Eight workers measure task time before and after a new tool.

Differences (after − before) average −2.4 minutes with s_d =
3.0. Test $\alpha = 0.10$ two-tailed.

- $t = -2.4 / (3.0/\sqrt{8}) \approx -2.26$, df = 7.
- Critical $t = \pm 1.895 \Rightarrow$ reject H_0.
- 90% CI: $-2.4 \pm 1.895 \cdot 3.0/\sqrt{8} \Rightarrow (-4.4, -0.4)$.
- Effect size $d = -0.80$.
- Conclusion: Tool significantly reduces task time.